



SECOND SEMESTER 2025-26
COURSE HANDOUT

Date: 06.01.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : **EEE/INSTR/ ECE F244**

Course Title : **Microelectronic Circuits**

Instructor-in-Charge : **ANU GUPTA**

Instructor(s) : **Neeraj Mishra,**

Tutorial/Practical Instructors: Neeraj Mishra, Nitin Chaturvedi, Ramesh Kumar, Govind Prasad, Prashant Kumar

1. Course Description: Basic microelectronic circuit analysis and design, biasing in discrete and integrated circuit amplifiers, an overview of modeling of microelectronic devices single and two transistor amplifier configurations with passive and active loads; current mirrors & current sources; single-ended and differential linear amplifiers , differential and multistage amplifiers; 2 stage CMOS OPAMP, frequency response of amplifiers; negative feedback in amplifiers, R-C frequency compensation.

2. Scope and Objective of the Course:

The objective of this course is to develop the ability to analyze and design integrated electronic circuits. The course aims to provide a thorough understanding of electronic circuits & building blocks necessary for the effective realization of analog integrated circuits. The course also includes the usage of SPICE as a circuit design aid.

3. Learning outcomes of the Course:

- Understand device modelling, two-port network models, and amplifier characterization parameters
- Understand the necessity of and techniques to set DC bias, Quiescent point location for different types of amplifiers.
- Understand analysis and synthesis of different single/ multi-stage amplifiers and their characterization.
- Design of amplifiers using EDA tools.

4. Text Book: Adel. S. Sedra, Kenneth C Smith, “Microelectronic Circuits”, Oxford University Press, Seventh Edition.

5. Reference Book/s:

Prime Reference Book/s: Behzad Razavi, “Design of Analog CMOS Integrated Circuits”, Tata McGraw-Hill, 2001.

Other reference books:

- a. Richard. C. Jaeger, “Microelectronic Circuit Design”, Tata McGraw-Hill Companies Inc., International Edition.
- b. R.Jacob.Baker, Harry.W.Li, David.Boyce, “CMOS circuit Design Layout and simulation. ”IEEE Press series on Microelectronic Systems, PHI.



6. Course Plan:

Approx. No. of Lectures	Module	Lecture Session	Reference From the Text Book/ reference book	Learning Outcomes Student will be able to --
1	Introduction --- Course handout, Review of circuit analysis techniques:	Review of circuit analysis techniques	Revision from previous courses	Revision of concepts of previous courses
2	Opamp / operational trans conductance amplifier Design—top-down analysis	Specifications--2-stage/ 3 stage circuit Example/s of integrated electronic systems— An overview	Filter, oscillator, signal generator, wave shaping circuits, tuned amplifiers	Derive Specifications for design Overview of electronic systems
3	Introduction to Amplifiers, basic configurations	Class-A amplifier, Basic performance measures of Amplifiers,) : VTC, quiescent point, power efficiency, gain, , amplifier Topologies,	Text chapter-1 - 1.4, 1.5, 1.6	Characterize an Amplifier's performance
5	Basic building blocks: MOSFET, operation regions, and small-signal models. Overview--Revision BJT devices operation	Models of MOSFET, operation and drain current characteristics (with voltages) of MOSFET. Overview (self-reading)	Text Ch- 5 Prime Ref -chapter 2 Text Ch- 6	Understand MOS device characteristics, models
10	DC Biasing and Bias point stability: Biasing using resistors, current mirrors (basic, cascode, low voltage cascode) Single-stage amplifier design Frequency response of amplifiers	Voltage biasing and current biasing--Passive and active current mirrors. Overview--- Discrete/ Integrated circuit MOSFET Amplifiers, BJT models, BJT Amplifiers (self-reading) Integrated circuit MOSFET Amplifier circuits, and Frequency response	Prime Ref —Ch. 3, 5 ch. 7, Text Ch. 8, Text Ch 7, 8, Ch. 7 (self-reading) Prime Ref Ch. 3, 6 Text Ch. 10	Design of a basic discrete MOSFET Amplifier Understanding of MOSFET-based IC Amplifiers Design and analysis techniques Designing of DC bias circuits
10	Differential Amplifier Multistage high-gain Operational Amplifiers	Differential amplifiers	Prime Ref -Ch. 4 Text --Ch.9	Designing differential amplifiers and multistage amplifiers



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		Operational Amplifiers design	Prime Ref ch-9 Text Ch.9	Design and characterization of an integrated OP-AMP
10	Feedback Amplifiers Feedback topologies Two Port Networks: Z,Y,H,G, Stability/ compensation	Impact of negative feedback on the performance of amplifiers	Prime Ref ch-7 Text Ch. 10	Apply Concept of feedback in feedback amplifier design, analysis
		Stability & frequency compensation in OP AMP, Noise (only qualitative)	Prime Ref ch.-10, 7 Text Ch-11	Apply techniques to stabilize the OpAmp in feedback mode.
		Configuration of Feedback Amplifiers	Text Ch-11	
1	OpAmp/ OTA design Applications overview	Overview ---Example/s of integrated electronic systems— An overview	Filter, oscillator, signal generator, wave shaping circuits, tuned amplifiers	Building of electronic systems
42 total				

7. Evaluation Scheme:

Component	Duration	Marks/ Weightage	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	75 (25%)	As per the timetable	CB/OB
Regular Quiz (Weekly Tutorial hour)	20 / 25 min. Regular	60 (20%)	Spread throughout the semester	OB (Best 6 quiz out of total 8 quiz)
SPICE tool based Assignments, test, and/ or quiz	Spread throughout the semester	60 (20%)	To be announced during the semester	OB
Comprehensive Examination	3 Hr.	105 (35%)	As per the timetable	OB/CB
		300		

Note--



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*Each quiz will be of 10 marks. A total of 8 quizzes will be conducted, out of which the best 6 will be considered for grading.

8. Chamber Consultation Hour: To be informed by the respective tutorial instructor

To contact IC , fix a time. Email--anug@pilani.bits-pilani.ac.in

9. Notices: All notices related to the course will be put on the **EEE** Notice board/ **NALANDA** (LMS-e portal).

10. Minimum performance requirement for a valid grade: A student should obtain 25% of the average of the class, or 10% of the course total marks, whichever is lower, to clear the course. If any student gets marks lower than the prescribed standard mentioned above, he/she may be awarded an NC report.

11. Make-up Policy: Make-up for mid-semester and comprehensive examination will be considered ONLY in genuine cases, and the IC decision will be final. Prior permission from the instructor-in-charge is necessary before the examination. (**Note: No make-up will be given for regular quizzes/, SPICE quiz/ test**).

12. Note (if any): Lecture class regularity is essential.

Instructor-in-charge
Course No. EEE/INSTR F244